

IRED Sudoku on AWS

What happened, what the evidence actually supports, and how to run the next experiment safely and reproducibly

Decision in one sentence

Do not resume the final checkpoint or reuse the eight-GPU launcher unchanged. Start a fresh, seeded, 50,000-update experiment with global batch 64, a separate development split, permanent best and milestone checkpoints, corrected metrics, and automatic divergence stops.

Best recorded proxy

95.3%

strictly valid boards at step 10k; exact-grid rate was not logged

Final checkpoint

0%

strictly valid boards; cell accuracy returned to chance

Gross p4d compute

\$824.41

45.198333 Spot hours; credits offset the recorded charge

Prepared from the complete AWS system journal, final checkpoints, corrected 1,000-puzzle SATNet evaluations, Cost Explorer records, the public code release, and the ICML 2024 paper.

Document control	Value
Current through	30 September 2026
Repository	tx1yz0/ired_code_release
Evidence commit	2f9e7c06ea1df3279fec71e30fc13749ffa6b710
Primary training region	AWS us-east-1
Status	Retrospective complete; next run not yet launched

Metric correction: 99.5534% blank-cell accuracy is not the same measurement as the paper's reported 99.4% puzzle result. The closest retained proxy is 95.3% strict board validity, and the direct exact-grid result at step 10,000 is unavailable.

How to use this playbook

Three audiences

Researchers should use Sections 3-7 to interpret the result and lock the experiment. Operators should use Sections 8-12 to provision, monitor, stop, and recover the run. Reviewers should use the evidence appendix to trace every important claim.

Non-negotiable changes before launch

- [] Fresh run directory and explicit run ID; never auto-resume the collapsed checkpoint.
- [] Global batch 64. On more than one process, use split_batches=True and verify per-rank batch sizes.
- [] Fixed 50,000-update primary endpoint for the paper-aligned run; no 1.3 million-update continuation.
- [] Separate development data for monitoring. Keep the paper's 1,000 SATNet test puzzles untouched until the locked final evaluation.
- [] Direct exact-grid solve rate added as the primary metric; strict validity and blank-cell accuracy reported separately.
- [] Best, milestone, and last-known-good checkpoints uploaded off-instance; no rolling two-file policy.
- [] Immediate nonfinite stop plus a sustained loss-excursion stop, with diagnostics uploaded before shutdown.
- [] Seed, environment, dataset hashes, command, git SHA, hardware, and sampling RNG recorded with every run.

Fast reading paths

Goal	Read
Understand the previous result	Sections 1-7 and Appendix A
Prepare code and protocol	Sections 8-10
Operate and monitor AWS	Sections 7 and 11-14
Review claims and sources	Appendices A-B

Table of contents

How to use this playbook	2
Non-negotiable changes before launch	2
Fast reading paths	2
Table of contents	3
1. Executive findings	5
What the evidence establishes	5
What remains unknown	5
Correct interpretation of the 10k result	5
2. Run timeline and operating record	6
Resource state after the audit	6
3. Performance history	7
4. Metric definitions and reporting rules	8
Evaluation protocol	8
Sanity baselines	8
5. Root-cause analysis	9
Causal chain	9
Contributing code and process gaps	9
Ruled out or unlikely	9
6. Paper, public release, and AWS run	10
Public-material inconsistencies to disclose	10
7. AWS hardware, cost, and operational lessons	11
Historical hardware	11
Cost record	11
What worked operationally	11
What to change	11
8. Recommended next-run plan	12
Target primary configuration	12
9. Required code changes before launch	13
Target command after those changes	13
DDP equivalence test before any scale-out	13
10. Experimental matrix	14
Promotion rule	14
Inference ablations after the primary result	14
Result table template	14

11. Monitoring, quality gates, and stop rules	15
Live dashboard	15
Stop-state machine	15
Pilot gates	15
12. Checkpoint, logging, and recovery design	16
Checkpoint classes	16
Checkpoint payload	16
Off-instance artifact layout	16
Recovery procedure	16
13. Preflight checklist	17
Experiment lock	17
Reliability	17
AWS and cost	17
14. Live-run and closeout checklist	18
During the run	18
Final evaluation	18
Closeout	18
Go / no-go decision template	18
Appendix A. Evidence ledger	19
Checkpoint audit facts	19
Loss chronology	19
Appendix B. Sources and claim boundaries	20
Primary external sources	20
Interpretation boundaries	20
Arithmetic used in this document	20

1. Executive findings

The model learned, then the experiment lost the useful state

The full journal shows a strong early model, a widening train/test generalization gap, and an abrupt numerical collapse after prolonged training. The final checkpoint answers a different question from the paper because the protocol, total batch, exposure, metric path, and checkpoint selection all differed.

What the evidence establishes

- At step 10,000, the corrected evaluator reported 99.5534% blank-cell accuracy, 95.3% strict Sudoku validity, and 98.7593% on the release's weak sum-based score across all 1,000 SATNet test boards.
- Training-batch metrics remained 100% through step 630,000 while strict test validity fell into the low 30s. This shows a large generalization gap consistent with severe overfitting.
- The sampled loss was 0.0146 at step 636,000, exceeded $1e20$ around step 636,102, and remained around $1e24$ - $1e25$. At step 640,000, blank-cell accuracy was near 1/9 and board validity was zero.
- The final checkpoint loaded correctly. Model, EMA, and optimizer tensors were finite. The final 1,000 update interval changed optimizer counters but not model or EMA parameters, so optimization was effectively frozen by the end.
- The dataset passed structural checks, clues match solutions, all labels are one-hot, EC2 health checks stayed normal, no Spot interruption occurred, and shutdown was orderly.

What remains unknown

- The step-10,000 direct exact-grid solve rate. The checkpoint is gone and the journal did not log direct target equality.
- The exact numerical trigger for the step-636k collapse. Gradient-level telemetry and forensic batches were not retained.
- Whether the paper's 99.4% is exactly the released strict consistency metric. The paper does not map its result to the release's three metrics.
- RRN performance, seed-to-seed variance, and the effect of the paper's stated 3x3 output convolution versus the release's 1x1 layer.

Correct interpretation of the 10k result

Strong evidence, not an unbiased paper result

The 95.3% strict-validity result is the closest retained proxy to puzzle success and is 4.1 percentage points below the paper's 99.4%. Repeatedly examining the paper's 1,000-board test set also makes step 10,000 a diagnostic observation, not a clean model-selection result. The next run must separate development and test roles.

2. Run timeline and operating record

1	26 Sep - local bring-up Apple Silicon/MPS training reached step 2,011 in 3h 47m. Checkpoints at 1,000 and 2,000 were saved. The corrected eight-board diagnostic later showed 72.7528% blank-cell accuracy at step 2,000, confirming useful early learning.
2	27 Sep - AWS approval and deployment P-class Spot quota was approved at 96 vCPUs. A p4d.24xlarge Spot request launched in us-east-1c. The first distributed start hit an unused-parameter DDP error; find_unused_parameters=True was enabled and the successful run followed.
3	27-29 Sep - continuous training Eight A100 40 GB GPUs ran for 45.198333 Spot instance-hours. The service had no restart cycle, EC2 status checks stayed healthy, and all eight GPUs were active. Validation ran every 10,000 steps.
4	Completion - operational success The process reached step 1,300,000, wrote model-1300.pt, returned exit code 0, and powered the instance off before the 56-hour absolute deadline.
5	Final validation A CPU validation attempt on m7i.xlarge was too slow. A g4dn.xlarge T4 completed a 256-board check: 11.1164% blank-cell accuracy and zero strict-valid or sum-score boards.
6	30 Sep - forensic recovery Spot capacity was unavailable for the stopped p4d after three attempts. With approved On-Demand fallback, a temporary snapshot, volume clone, and t3.xlarge hosts recovered the journal and audited final checkpoints. All temporary audit resources and the SSH rule were deleted afterward.

Resource state after the audit

Resource	Identifier	State / lesson
Training instance	i-04da43a860b64ad85	Stopped. Do not rely on restarting it for recovery or validation.
Spot request	sir-yr3zm7kj	Normal shutdown disabled automatic continuation until explicitly restarted.
Training root volume	vol-025b2a9906642e9cb	100 GB gp3 remains attached and can continue accruing storage cost.
Temporary audit assets	snapshot, clone, two t3 hosts, SSH rule	Deleted after evidence recovery; original volume untouched.
Repository state	main @ 2f9e7c0	Contains the corrected history CSV, chart, run report, and AWS notes.

3. Performance history

The chart below is reconstructed from the complete system journal. Each validation point uses the corrected batch evaluator over all 1,000 SATNet boards with one stochastic sample per puzzle. The curve includes sampling noise, but the sustained decline and collapse are much larger than that noise.

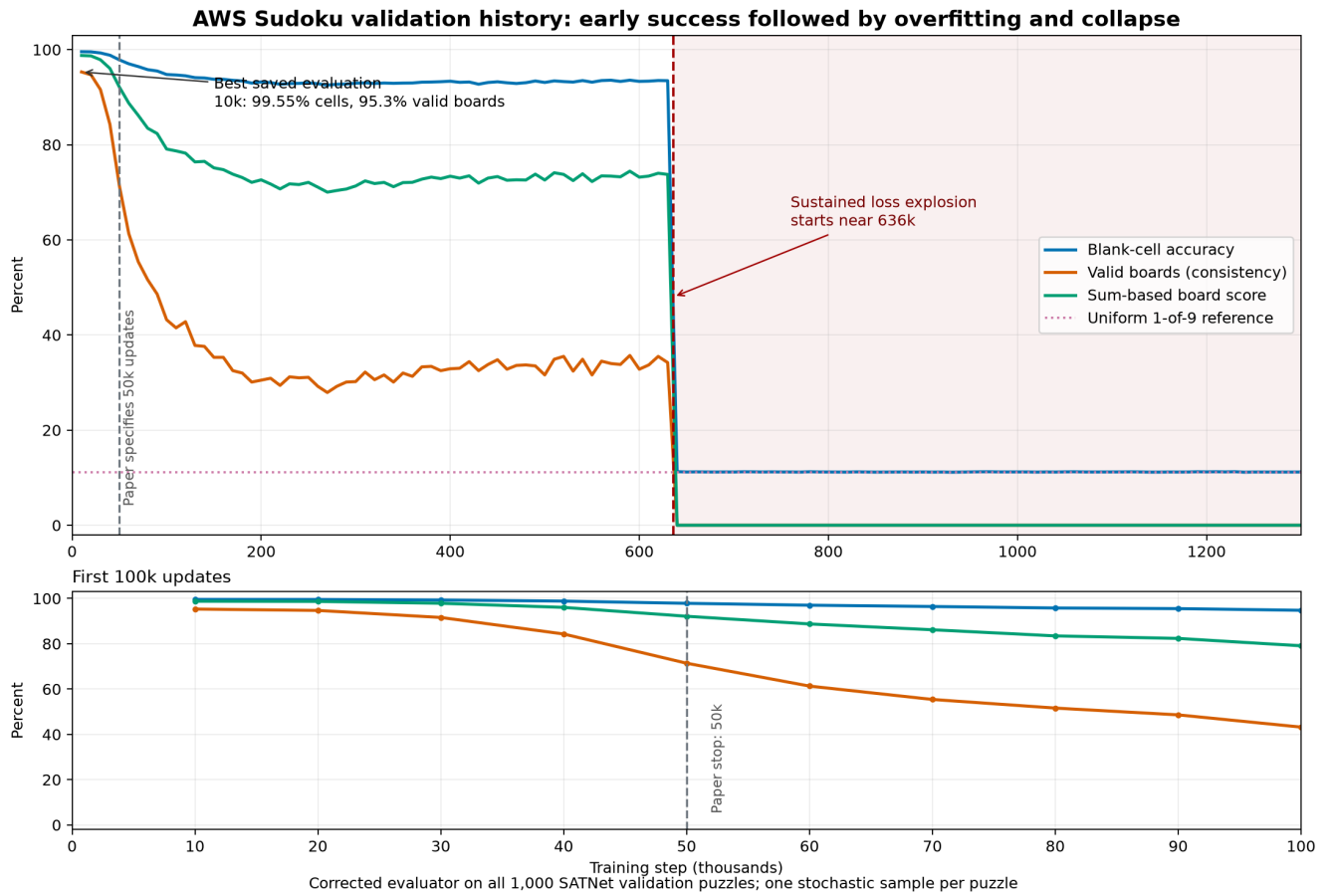


Figure 1. Full validation history and the first 100,000 updates. The paper's 50,000-update mark is an update count, not an exposure-equivalent point for the AWS batch-512 run.

Step	Blank cells	Strict validity	Sum score	Interpretation
10,000	99.5534%	95.3%	98.7593%	Best recorded diagnostic
20,000	99.5244%	94.7%	98.6790%	Early decline
50,000	97.8427%	71.4%	92.1309%	Paper update count; 7.72x its exposure
100,000	94.7650%	43.2%	79.0889%	Large generalization gap
630,000	93.4651%	34.2%	73.7494%	Last validation before collapse
640,000	11.2631%	0%	0%	Chance-level collapse
1,300,000	11.1937%	0%	0%	No recovery

4. Metric definitions and reporting rules

Primary metric for the next run

Use direct exact-grid solve rate: every predicted digit must equal the target. Report strict Sudoku validity, blank-cell accuracy, and clue preservation separately. Never use the release's board_accuracy as the headline result.

Metric	What it measures	Use	Known limitation
Exact-grid solve rate	Fraction of boards where all 81 predicted digits equal the target	Primary paper-comparison metric	Not present in the historical journal; must be added
Strict consistency	Rows, columns, and 3x3 boxes each contain unique digits	Primary validity diagnostic	Does not directly compare with target; likely close when clues are clamped and solutions are unique
Blank-cell accuracy	Correct digits only at originally empty cells	Learning signal and secondary metric	A high value can coexist with many invalid full boards
Clue preservation	Given cells remain unchanged	Required invariant	Must be logged explicitly even though the sampler clamps clues
board_accuracy	Per-cell check that row/column/box sums equal 36	Legacy diagnostic only	Duplicate digits can satisfy sums; neither exact solve nor strict validity
Training loss	Denosing and energy objective	Health signal	Low loss did not guarantee held-out validity; loss later exploded

Evaluation protocol

- For the pilot, carve development data from the 9,000 training boards for diagnostics and stop-rule design. Keep the paper's 1,000 SATNet test boards untouched. After the protocol is locked, retrain the primary paper-aligned condition on all 9,000 training boards.
- Lock architecture, seed set, and inference settings before the primary run. Use its fixed 50,000-update endpoint without selecting on the test set. A development-best checkpoint from the pilot is a secondary operational result, not the paper comparison.
- Use a predeclared sampling seed and one sample per puzzle for the primary result. Report any best-of-k or additional-compute result separately.
- Run all 1,000 SATNet test boards once per trained seed. Run the 18,000 RRN boards only after configuration selection.
- Report mean and standard deviation across at least three training seeds, plus per-seed exact-grid, strict-validity, blank-cell, and inference-time results.

Sanity baselines

Input	Expected blank-cell result	Expected board result
Ground-truth label	100%	100% exact-grid, strict validity, and clue preservation
Uniform digit guess	About 11.111%	Approximately 0% exact or strict-valid boards
Constant digit	Near class frequency	0% strict-valid boards
Historical final EMA	11.1164% on 256-board confirmation	0% strict validity

5. Root-cause analysis

Causal chain

A	Protocol drift The paper describes 50,000 updates and global batch 64. AWS used 1,300,000 updates and global batch 512 at the same learning rate.
B	Comparable exposure arrived early The paper's 3.2 million sample presentations were reached near total step 8,000. Step 10,000 used about 4.224 million presentations.
C	Generalization degraded Training metrics stayed saturated while held-out strict validity fell from 95.3% to the low 30s. This is consistent with severe overfitting, though no controlled ablation isolates the batch and schedule effects.
D	Numerical collapse A sustained loss explosion began around step 636,102. The next validation was at chance, and the model never recovered.
E	Guardrails did not stop it No best-checkpoint pin, divergence threshold, nonfinite guard, scheduler, or model-quality exit condition was active.
F	Retention erased recovery points The rolling two-checkpoint policy preserved only collapsed final states and deleted the useful early checkpoints.

Contributing code and process gaps

- aws/run_sudoku.sh explicitly set --split-batches False on eight processes, so batch_size 64 became global batch 512.
- The launcher treated process exit 0 as success and powered off without checking model quality.
- The service could restart deterministic failures indefinitely because Restart=on-failure had no bounded retry classification.
- Validation was only every 10,000 steps, metrics were partly ambiguous, and no direct exact-grid rate was logged.
- Logs and checkpoints lived primarily on the root volume rather than continuously in CloudWatch and S3.
- The public release provides no seed, environment lock, tagged experiment config, pretrained Sudoku checkpoint, or variance report.

Ruled out or unlikely

Candidate	Evidence	Assessment
Bad SATNet data	10,000 finite one-hot boards; 9k/1k split; 31-42 clues; all clues match targets	Ruled out as the primary cause
Spot interruption	One continuous boot, no interruption event, orderly shutdown	Ruled out
EC2 hardware failure	StatusCheckFailed stayed zero; GPUs active	No supporting evidence
Corrupt final file	Both final files loaded; all model/EMA/optimizer tensors finite	Ruled out
Immediate DDP failure	First launch failed, then the patched run trained continuously	Operational incident, not the later collapse
Final evaluator bug	Fork uses samples, not samples[-1]; 256-board confirmation agrees with journal	Historical public bug corrected

6. Paper, public release, and AWS run

Dimension	Paper	Public release	AWS run
Training updates	50,000	train.py hardcodes 1,300,000	1,300,000
Hardware	One RTX 2080	Not fixed	Eight A100 40 GB
Global batch	64	Trainer default split_batches=True	512 via 64 per process and split_batches=False
Learning rate	Adam 1e-4	1e-4	1e-4
Sample presentations	3.2 million	Depends on process semantics	664.704 million
Final convolution	3x3 Conv2D to 9	1x1 Conv2D to 9	1x1 release model
Validation call	Not specified	samples[-1] broadcast against batch	Corrected full samples tensor
Primary metric	Reported 99.4%; mapping unclear	Three differently scoped metrics	Exact-grid not logged; strict proxy peaked 95.3%
Seeds / variance	Not reported	Not provided	Single training trajectory
RRN test	Full 18,000	Implemented	Skipped

Update multiplier 26x 1.3m / 50k	Batch multiplier 8x 512 / 64	Exposure multiplier 207.72x 664.704m / 3.2m
--	--	---

Public-material inconsistencies to disclose

- The paper's 50,000-update protocol conflicts with the release's 1.3 million-update default.
- The paper's final 3x3 convolution conflicts with the release's 1x1 layer.
- The public validation path evaluates one prediction against a full label batch through broadcasting.
- Paper Algorithm 2 describes repeated energy-gradient optimization, while the release sampler also performs a diffusion p_sample update before each set of 20 refinements.
- The metric named board_accuracy is a weak sum test and must not be interpreted as boards solved.

Reproduction claim boundary

An exact reproduction is impossible from public artifacts alone. The next run can be a defensible paper-aligned reproduction attempt if it declares which specification it follows, freezes the endpoint and metrics in advance, and reports all deviations.

7. AWS hardware, cost, and operational lessons

Historical hardware

Workload	Instance	Purpose / outcome
Main training	p4d.24xlarge Spot - 8x A100 40 GB, 96 vCPU, 1,152 GiB	45.198333 hours; completed 1.3m updates
CPU validation	m7i.xlarge On-Demand	Too slow for practical full sampling
GPU validation	g4dn.xlarge On-Demand - 1x T4 16 GB	Completed 256-board final check
Forensic audit	Two t3.xlarge On-Demand hosts	Recovered journal and audited checkpoints

Cost record

Item	Usage	Gross	Credits / note
p4d Spot compute	45.198333 hours	\$824.41	Recorded credit: -\$824.41
gp3 EBS in run window	6.666667 GB-month	\$0.53	Account-scoped; recorded credit: -\$0.53
Public IPv4 in run window	46.2025 hours	\$0.23	Account-scoped; recorded credit: -\$0.23
m7i + g4dn validation	36m38s + 10m39s	About \$0.22	Estimate; billing lagged
t3 forensic hosts	24m04s combined	About \$0.07	Estimate; temporary storage extra

24-hour g5.xlarge planning reference - queried 30 Sep 2026

Linux On-Demand was \$1.006/hour, or \$24.14 for 24 hours. Recent us-east-1 Spot observations ranged from \$0.3986 to \$0.6035/hour, or \$9.57-\$14.48 for 24 hours. A \$0.65/hour Spot ceiling is \$15.60 for 24 hours. These are compute-only gross estimates before credits; capacity and price can change.

Credits are not a cost model

The recorded p4d charges were offset by credits, but future runs should be approved against gross cost. The retained 100 GB root volume can keep billing after compute stops, and Cost Explorer data can arrive late.

What worked operationally

- An absolute systemd shutdown deadline survived stop/restart cycles and bounded the compute window.
- Atomic checkpoint writes prevented partial final files.
- The systemd service supported restart after transient failure and powered off after process completion.
- The P-class Spot quota request was approved before the main deployment.

What to change

- Use a single-GPU pilot first. The paper used one RTX 2080; a g5.xlarge A10G pilot offers 24 GB memory and avoids paying for eight A100s before correctness is established. Start with a 24-hour absolute ceiling and verify fit and throughput with a short benchmark before committing.
- For any later Spot scale-out, use multiple Availability Zones and capacity-optimized allocation. A stopped p4d is tied to one zone and may be impossible to restart when capacity is scarce.
- Stream metrics and journal logs to CloudWatch and checkpoints to S3. Recovery must not depend on mounting a stopped root volume.
- Tag every EC2, EBS, snapshot, S3, and log resource with project, run ID, seed, owner, and expiry. Enable cost allocation tags.
- Set 50%, 80%, and 100% budget alerts on gross projected cost. Require an explicit ceiling for any On-Demand fallback.
- Terminate compute after artifact upload. Snapshot or delete root storage according to the retention policy, then verify no temporary security rule remains.

8. Recommended next-run plan

Recommended sequence

Fix metrics and guardrails, run a cheap single-GPU pilot, validate distributed equivalence only if scale is needed, then execute the locked 50,000-update experiment. Do not schedule another 1.3 million-update run.

0

Freeze the protocol

Choose paper 3x3 or release 1x1 as an explicitly named condition. Predeclare global batch 64, Adam 1e-4, 50k endpoint, seeds, metrics, dev/test split, sampler settings, and stop rules.

1

Metric and data sanity

Ground truth must score 100% on all metrics. Random and constant predictions must produce expected baselines. Hash datasets and record the exact split.

2

Single-GPU smoke test

Run 100 steps, reload the checkpoint in a fresh process, verify all tensors are finite, reproduce fixed-seed metrics, confirm log and S3 uploads, and trigger a deliberate guard in a test mode.

3

Gated 10k pilot

Use one GPU and the development split. Evaluate at 1k, 2k, 5k, and 10k. The historical curve suggests strong learning by 10k, but do not tune on the 1,000-board test set.

4

Lock and run 50k

After the pilot passes, freeze the configuration. Retrain from scratch on all 9,000 training boards to the fixed 50k endpoint. Preserve pinned milestones, last-known-good, and final checkpoints off-instance; do not select on the SATNet test set.

5

Repeat seeds

Run at least three training seeds. Do not promote a result based on a single favorable trajectory.

6

Final evaluation

Evaluate each locked 50k endpoint once on all 1,000 SATNet test boards. Report exact-grid solve rate first. Run RRN only after the configuration is selected.

Target primary configuration

Parameter	Locked value	Reason
Training start	Fresh initialization	Final checkpoint is collapsed; auto-resume is unsafe
Device count	1 for primary paper-aligned run	Matches the paper's single-device semantics and is cheapest
Global batch	64	Paper setting; if distributed, split across ranks
Optimizer	Adam, learning rate 1e-4	Paper and release setting
Updates	50,000 fixed primary endpoint	Paper setting; prevents schedule drift
Landscapes	10 cosine diffusion steps	Release and historical setting
Inner refinement	20 steps per landscape	Release and historical setting
EMA	0.995, update every 10	Current release configuration; report raw model diagnostically
Validation	Pilot dev split; primary uses fixed endpoint	Tune guardrails before lock, then train the paper-aligned condition on all 9k
Test	Untouched SATNet 1,000 once per seed	Unbiased final comparison

9. Required code changes before launch

Priority	Change	Current location	Acceptance check
P0	Change split_batches to True; compute and print global batch	aws/run_sudoku.sh:15-29	One GPU sees 64; eight GPUs see 8 each and global 64
P0	Remove implicit latest resume; require run ID and explicit resume path	aws/run_sudoku.sh:31-34	Fresh run cannot load an old directory accidentally
P0	Add seed and persist all RNG states	train.py + checkpoint save/load	Fresh reload reproduces the next batch and sample
P0	Add exact-grid and clue-preservation metrics; fix blank-cell aggregation	diffusion....:1108-1111, 1230+	Metric unit fixtures pass
P0	Create dev/test separation	sat_dataset.py + train.py	Test loader is inaccessible during training
P0	Add nonfinite and sustained loss-excursion stops	diffusion....:977-989	Injected NaN and huge-loss tests stop with diagnostic exit code
P0	Pin best, milestones, and last-known-good; upload before prune	diffusion....:881-908	1k/2k/5k/10k/50k plus best and last-good survive
P1	Log gradient norm, parameter norm, EMA distance, energy margin, entropy	training step around clip at :984	Dashboard shows each signal by step
P1	Synchronize all ranks around rank-0 evaluation	diffusion....:997-1010	No rank advances while validation is active
P1	Bound restart behavior by exit-code class	aws/bootstrap_sudoku.sh	Quality failure uploads and stops; transient interruption resumes
P1	Add CloudWatch/S3 persistence and completion manifest	bootstrap + run launcher	Host termination leaves complete logs, metrics, config, and hashes

Target command after those changes

The current launcher is historical and should not be run unchanged. After adding --seed, --run-id, --dev-split, and explicit artifact destinations, the primary shape should be equivalent to:

```

accelerate launch --num_processes 1 train.py \
  --dataset sudoku --model sudoku --batch_size 64 \
  --train-steps 50000 --split-batches True \
  --cond_mask True --supervise-energy-landscape True \
  --use-innerloop-opt True --diffusion_steps 10 \
  --data-workers 4 --evaluate-every 1000 \
  --skip-rnn-test --seed <LOCKED_SEED> \
  --run-id <UNIQUE_RUN_ID> --resume-from none
  
```

Checkpoint storage sizing

Historical checkpoints are about 701 MiB each. Keeping every 1,000-step file for a 50,000-step run is roughly 35 GiB before other artifacts. A practical policy keeps best checkpoints, pinned milestones, and a rolling recent window locally while uploading every gated artifact to S3 with a checksum.

DDP equivalence test before any scale-out

- 1 Start from identical weights and a fixed 64-example batch.
- 2 Run one optimizer update on one GPU with batch 64.
- 3 Run the same update on eight GPUs with split_batches=True and eight examples per rank.
- 4 Compare total and component losses, pre-clip gradient norm, parameter delta, and EMA update within declared numeric tolerance.
- 5 Repeat for ten updates before authorizing distributed training.

10. Experimental matrix

The public paper and code conflict. Resolve that conflict with named conditions instead of silently combining them. Run the cheapest, most informative condition first and expand only after it passes.

ID	Purpose	Output conv	Devices	Global batch	Updates	Seeds	Decision
A	Paper-described baseline	3x3	1	64	50k	3-5	Closest textual reproduction
B	Public-release baseline	1x1	1	64	50k	3-5	Closest executable reproduction
C	DDP equivalence	1x1	8	64 split	50k	3	Run only after 1-vs-8 equivalence passes
D	Batch ablation	1x1	8	512	50k cap	3	Explains prior run; separate from reproduction

Promotion rule

Start with B, then A

Condition B requires the fewest code changes and tests the public executable specification. If it passes metric and stability gates, run A to resolve the 3x3-versus-1x1 paper conflict. Do not use conditions C or D to claim a paper reproduction.

Inference ablations after the primary result

- Refinement steps T in {5, 10, 15, 20}. Keep T=20 as the locked primary result; label extra compute clearly.
- EMA versus online weights as a diagnostic, without selecting whichever looks best on test.
- One sample per puzzle as primary; multi-sample or best-of-k only as separate analysis.
- Optional warmup or cosine learning-rate decay only after the fixed-1e-4 reproduction, as a stability ablation.

Result table template

Condition	Seed	Exact solve	Strict valid	Blank cells	Clues	RRN	Cost
B - release 1x1		<%>	<%>	<%>	<%>	deferred	\$
B - release 1x1	mean +/- std	<%>	<%>	<%>	<%>	<%>	\$
A - paper 3x3		<%>	<%>	<%>	<%>	deferred	\$

11. Monitoring, quality gates, and stop rules

Live dashboard

Signal	Cadence	Alert / stop behavior
Total, denoising, and energy loss	Every step; aggregate 10/100/1000	Immediate stop on nonfinite. Diagnostic pause on sustained >100x trailing median or any extreme excursion.
Pre-clip grad norm and clipped fraction	Every step	Stop on nonfinite; alert on sustained saturation at clip 1.0
Parameter norm and EMA-online distance	Every 100 steps	Alert on abrupt jump, collapse to zero, or no movement while optimizer advances
Dev exact solve / strict validity / blank cells	1k, 2k, 5k, 10k; then every 5k or 10k	Pin best; stop pilot on sustained regression according to the predeclared rule
Digit entropy and class histogram	Every validation	Alert on uniform chance or single-class collapse
Clue preservation	Every validation	Immediate quality failure if below 100%
Step rate and checkpoint age	Every minute	Alert if stalled; recompute completion and gross cost projection
GPU memory/utilization, NCCL, EC2 health	CloudWatch at 1 minute	Transient infrastructure failure may resume; deterministic quality failure must not restart
Spot interruption notice	Continuous	Checkpoint, upload, flush logs, exit cleanly

Stop-state machine

State	Trigger	Required action
GREEN	Finite values; dev trend within declared range	Continue; upload scheduled artifacts
AMBER	Sustained regression, excessive clipping, throughput/cost drift	Pin checkpoint; upload forensic batch; pause for review
RED - quality	NaN/Inf, clue violation, huge sustained loss, frozen parameters	Save last-known-good and diagnostics; exit non-retryable; stop instance
RED - infrastructure	Spot notice, disk pressure, host/NCCL failure	Atomic checkpoint; upload; exit retryable; resume only from verified file
COMPLETE	Locked 50k endpoint and artifact manifest verified	Upload final; evaluate from a fresh process; terminate compute

Pilot gates

- Step 0: metric fixtures pass; dataset hashes match; random baseline is near 11.1% blank-cell accuracy and near 0% strict validity.
- Step 100: checkpoint reload reproduces fixed-seed outputs; logs and artifacts are visible off-instance.
- Steps 1k/2k/5k/10k: dev metrics show learning, all tensors remain finite, and no clue violation occurs. The prior run suggests the model should be strong by 10k, but thresholds must be declared before observing the new dev results.
- At any new dev best: write an immutable best checkpoint and verify it in a fresh evaluator before pruning anything.
- At 50k: finish even if a dev peak occurred earlier, unless a safety guard fired. This preserves the paper's fixed endpoint as the primary comparison.

Exit code 0 is not model success

Completion requires a verified final manifest and a quality report. The launcher must not power off merely because the training process returned zero; it should first upload artifacts and record whether quality gates passed.

12. Checkpoint, logging, and recovery design

Checkpoint classes

Class	Examples	Retention	Verification
Best dev (pilot only)	best-exact, best-strict, best-blank	Immutable for the pilot	Reload, finite scan, fixed-seed evaluation
Pinned milestone	1k, 2k, 5k, 10k, 20k, 30k, 40k, 50k	Project retention policy	Checksum + manifest
Recent rolling	Last 3-5 checkpoints	Until next verified save	Step in filename equals stored step
Last known good	Most recent green-state save	Until run closeout	No nonfinite tensors; quality state attached
Forensic	Guard-trigger batch, RNG, grads/norms, config	Until incident review	Linked to exact failing step

Checkpoint payload

- Online model, optimizer, EMA, AMP scaler, step, and optional scheduler.
- Python, NumPy, CPU Torch, and all CUDA RNG states; sampling seed stream and DataLoader state.
- Resolved configuration; run ID; seed; git SHA; dirty flag; exact command; world size; rank/local/global batch.
- Python/package lock, CUDA, driver, instance type, GPU model, TF32/determinism settings, and dataset hashes.
- Best-so-far metrics, quality state, timestamp, parent checkpoint, and content checksum.

Off-instance artifact layout

```
s3://<bucket>/ired-sudoku/<run-id>/
config/resolved.json
environment/{git.txt,packages.txt,cuda.txt,dataset_hashes.json}
metrics/{train.jsonl,dev.jsonl,final_test.json}
checkpoints/{best/,milestones/,recent/,last-good/}
diagnostics/<step>/{batch.pt,health.json,logs.txt}
completion/manifest.json
```

Recovery procedure

- 1 Provision a clean host from the pinned image or container; do not reuse unknown local state.
- 2 Download one explicit checkpoint by checksum. Never resolve 'latest' from a shared folder.
- 3 Reload in a fresh process, assert all tensors finite, confirm stored step/config, and rerun fixed-seed dev metrics.
- 4 Resume only after metrics match the checkpoint manifest within tolerance. Record the resume event and new host identity.
- 5 For a quality/divergence exit, do not auto-resume. Diagnose from last-known-good plus the forensic bundle.

13. Preflight checklist

Launch authority

The operator signs this page only after the code, data, budget, and recovery path have been independently verified.

Experiment lock

- [] Condition ID, architecture (1x1 or 3x3), seed set, endpoint, global batch, learning rate, dev/test split, sampler settings, and primary metric are written in the run config.
- [] The run starts from fresh initialization in a unique output directory.
- [] Global batch is printed and asserted as 64. Per-rank batch is correct for the selected world size.
- [] Paper test data is inaccessible from the training and dev-evaluation path.
- [] Metric fixtures pass for labels, random guesses, constant predictions, and clue preservation.
- [] One-GPU versus DDP equivalence has passed if more than one GPU will be used.

Reliability

- [] Checkpoint round-trip, checksum, RNG restoration, finite scan, and fixed-seed evaluator pass in a fresh process.
- [] Best, milestone, rolling, last-good, and forensic retention classes are configured.
- [] Nonfinite and sustained loss-excursion guards have been deliberately triggered in a test mode.
- [] CloudWatch receives logs and health metrics; S3 receives checkpoints and JSON metrics.
- [] Restart policy distinguishes retryable infrastructure exits from non-retryable quality exits.
- [] Disk-space alarm accounts for checkpoint size and upload lag.

AWS and cost

- [] Current On-Demand and Spot rates were re-queried; a short benchmark produced measured steps/hour and validation overhead.
- [] Gross cost ceiling equals measured projected hours x live rate plus 25% headroom, EBS, snapshots, logs, IPv4, and final evaluation.
- [] Budget alerts are active at 50%, 80%, and 100%; On-Demand fallback has an explicit ceiling.
- [] Quota and capacity are confirmed in more than one Availability Zone where possible.
- [] IAM role grants only required S3/CloudWatch actions; IMDSv2 and restricted network access are configured.
- [] Instance, volume, snapshot, bucket prefix, and log group carry project/run/seed/expiry tags.
- [] The 24-hour absolute shutdown deadline, completion upload, and cleanup procedure have been dry-run.

Approval	Name	Date / time	Evidence
Research protocol			Config hash:
Code and metrics			Commit:
AWS and budget			Ceiling:
Launch operator			Run ID:

14. Live-run and closeout checklist

During the run

- [] Confirm step rate, projected finish, and gross projected cost after the first 100 and 1,000 steps.
- [] At every gate, review dev exact-grid, strict validity, blank cells, clues, loss components, gradient norm, EMA distance, entropy, and throughput together.
- [] Verify immutable best/milestone upload and checksum before any local pruning.
- [] Investigate every AMBER state. Do not normalize a repeated alert without updating the predeclared protocol and creating a new run ID.
- [] On Spot notice or infrastructure failure, checkpoint and upload before exit. On quality failure, stop without automatic retry.
- [] Do not inspect the 1,000-board SATNet test set during training or checkpoint selection.

Final evaluation

- [] The fixed 50k checkpoint is reloaded on a clean evaluator and passes finite/checksum/config checks.
- [] One predeclared sample per puzzle is run on all 1,000 SATNet test boards with the locked seed.
- [] Exact-grid solve rate is reported first, followed by strict validity, blank-cell accuracy, clue preservation, inference time, and legacy sum score.
- [] All training seeds are reported, including failures; mean and standard deviation are calculated deterministically.
- [] RRN 18k is run only after configuration selection and is reported as a separate out-of-distribution result.

Closeout

- [] Upload final test JSON, configuration, environment, logs, checkpoints, plots, and completion manifest.
- [] Record gross and net AWS cost with query timestamp and attribution limits.
- [] Terminate compute; delete or intentionally retain EBS/snapshots; remove temporary ingress; verify resource inventory is empty except approved retention.
- [] Commit the run report and artifact hashes to the repository.
- [] Write a one-page decision: reproduce, investigate, or stop. Include the evidence threshold for the next spend approval.

Go / no-go decision template

Decision	Minimum evidence	Action
GO - repeat seeds	Pilot metrics and stability pass; artifacts reproducible; projected gross cost within ceiling	Run remaining locked seeds
GO - final test	All seeds reach fixed endpoint; no protocol changes after lock	Evaluate untouched SATNet 1k
PAUSE	Learning occurs but metrics regress, vary widely, or miss threshold	Analyze dev-only evidence; revise as a new condition
STOP	Nonfinite/collapse, no learning, clue violation, irreproducible reload, or budget breach	Preserve forensics and terminate resources

Appendix A. Evidence ledger

Evidence	Location	Supports
Consolidated run report	AWS_SUDOKU_RUN_REPORT.md	Performance, cost, hardware, root cause, checkpoint audit
Recovered validation history	aws/sudoku_validation_history.csv	All 130 test evaluations from 10k through 1.3m
Validation chart	aws/sudoku_validation_curve.png	Early peak, generalization gap, collapse
Historical launcher	aws/run_sudoku.sh	Batch512, 1.3m updates, cadence, retention
Bootstrap and deadline	aws/bootstrap_sudoku.sh; aws/spot_deadline_user_data.sh	Service restart and absolute shutdown behavior
Local report	REPORT.md	MPS bring-up, step2k checkpoint, initial evaluator defect
Trainer	train.py; diffusion_lib/denoising_diffusion_pytorch_1d.py	LR, EMA, DDP, save/load, clipping, validation, metrics
Dataset loader	sat_dataset.py	SATNet split and RRN ingestion

Checkpoint audit facts

- model-1299.pt and model-1300.pt were each 735,313,035 bytes.
- Online model, EMA, and optimizer contained zero NaN or Inf values in both files.
- All 68 online model parameter tensors and all 68 EMA parameter tensors were exactly equal across the final 1,000 steps.
- Online versus EMA RMS difference at the end was approximately 4.15e-7; max difference approximately 4.77e-5.
- The audit host used a cloned volume; the original training volume was not modified.

Loss chronology

Step / range	Observation
239,804	First isolated recorded loss above 1; later recovered
381,827	First isolated recorded loss above 1e6; later recovered
596,803	First isolated recorded loss above 1e12; later recovered
636,000	Sampled total loss 0.0146; denoising 0.0005; energy 0.2829
Around 636,102	First sustained value above 1e20
637,000	Loss about 8.90e24
1,112,105	Maximum observed journal loss about 2.44e25
Final	Loss remained around 1e24-1e25; parameters no longer changed at representable precision

